Snekhal Prasad

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SUMMARY

NIT Trichy graduate and Software Engineer with ~2 years of experience in AI inference optimization across GPUs and edge AI accelerators, accelerator backend development, and graph and memory optimizations, with additional experience building agentic AI applications.

EXPERIENCE

Oracle — Jan 2026 – Oct 2026

Applications Developer - 1

Developed an end-to-end **agentic AI** application for enterprise approval automation across procurement, access, finance, and HR workflows.

- Built a policy evaluation framework using JSON rules and LLMs reconciling both outputs to evaluate **200+** enterprise policies through token-budgeted parallel inference and process supporting documents.
- Orchestrated **multi-agent** workflows using **LangGraph** StateGraph, shared graph state, and conditional routing across retrieval, validation, and decisioning, reducing approval processing time by **~80%**.
- Implemented a **RAG pipeline** for document-grounded conversational AI system using **LangChain** through chunking and embedding vector search to retrieve context with AI guardrails across **100**+ enterprise documents.

Blaize

Software Engineer I — **May 2025 – Jan 2026**

Integrated **Llama.cpp** with Graph Streaming Processor (GPU), enabling **accelerated** LLM inference on edge devices.

- Implemented **GPU execution paths** for compute-intensive operations, including **MatMul/GEMM**, handling tensor layouts and data movement offloading **45%** of AI compute workloads for models up to 7B parameters.
- Improved inference time by **26%** through **graph reuse** and **quantization** optimizations.
- Built a tensor and **KV-cache** memory reuse layer, reducing redundant memory allocations by **75%** through caching and buffer reuse.
- **Profiled** inference workloads to identify execution bottlenecks and evaluate operator-level latency, graph execution overhead, and memory utilization.

Engineering Intern — **Dec 2024 – May 2025**

Contributed to a facial-recognition attendance system deployed at Blaize Hyderabad, serving **200+** employees.

- Profiled ResNet50 inference on target hardware and optimized execution through image batching and **OpenCL and assembly kernel modifications**, improving inference throughput by **75**%.
- Built an AI chatbot backend using **LlamaIndex** for natural-language querying of SQL data, reducing lookup time by 72% with up to 84% response accuracy.

EDUCATION

National Institute of Technology, Tiruchirappalli — **June 2021 – May 2025**

BTECH, Metallurgical and Materials Engineering — CGPA: 8.02

ACHIEVEMENTS

- Received Blaize's **Above & Beyond** Award for contributions to engineering initiatives.
- Earned LeetCode's Knight Badge, placing among the **top 5%** of users. (leetcode.com/lahkens)

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, JavaScript, TypeScript
- **AI/ML & Systems:** LLMs, Generative AI, Agentic AI, RAG, LangChain, LangGraph
- **Frameworks & Technologies:** Llama.cpp, OpenVX, OpenCL, React.js, FastAPI, REST APIs, Git, Docker, GitLab